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Getting-off assistance device for a vehicle

Abstract

A getting-off assistance device that includes a surrounding information acquisition device that acquires information about the surroundings of a vehicle and a control ECU that controls a notification device that notifies an alert to an occupant or occupants of the vehicle, and the control ECU is configured to start the notification of the alert when it is determined that it is dangerous for the occupant to get off the vehicle based on information acquired by the surrounding information acquisition device, end the notification of the alert when it is determined that it is no longer dangerous for the occupant to get off the vehicle, and change the alert so as to reduce annoyance of the alert given to the occupant or occupants when the number of times of the alerts after the door is finally opened is equal to or greater than a reference number of times.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Japanese Patent Application No. JP2021-212742 filed on Dec. 27, 2021, the content of which is hereby incorporated by reference in its entirety into this application.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a getting-off assistance device for a vehicle such as an automobile.

2. Description of the Related Art

(3) A getting-off assistance device for a vehicle such as an automobile has been known, which is configured to operate an alert notification device to notify an occupant or occupants of an alert when a door is opened and it is determined that it is dangerous for the occupant or occupants to get off the vehicle. For example, in Japanese Patent Application Laid-open No. 2014-085869, a getting-off assistance device is described that notifies an occupant or occupants of information about approaching of another vehicle by means of a speaker or the like when there is another vehicle approaching an own vehicle on the rear side of the own vehicle and a door of the own vehicle on the side where the other vehicle is approaching is open.

(4) According to this type of getting-off assistance device, it is possible to improve the safety of the occupant or occupants when getting off the vehicle, compared to where the occupant or occupants are not notified of warning.

(5) However, in the above-described conventional getting-off assist device, an alert is issued every time an object such as another vehicle approaches the own vehicle while a door of the own vehicle is open, so that in a situation where a plurality of vehicles approach and pass one after another, an occupant or occupants may feel annoyed by repeated alerts.

SUMMARY

(6) The present disclosure provides an improved getting-off assistance device which can notify an occupant or occupants of an alert when getting off the vehicle is dangerous, and is improved so as to reduce the possibility that an occupant or occupants feel annoyed by alerts in a situation where a plurality of other vehicles approach and pass one after another.

(7) The present disclosure provides a getting-off assistance device for a vehicle that includes an surrounding information acquisition device that acquires information about surroundings of an own vehicle, a notification device that notifies an alert to an occupant or occupants of the own vehicle, an open-door detection device that detects opening of a door, and an electronic control unit that controls the notification device, and the electronic control unit is configured to start the notification of the alert by the notification device when it is determined that it is dangerous for the occupant to get off the vehicle based on the information acquired by the surrounding information acquisition device in a situation where the open-door detection device detects that a door is open, and end the notification of the alert by the notification device when it is determined that it is no longer dangerous for the occupant to get off the vehicle based on the information acquired by the surrounding information acquisition device.

(8) The electronic control unit is further configured to change the alert notified by the notification device so as to reduce annoyance of the warning given to the occupant or occupants when a number of times of the alerts notified by the notification device after the door is finally opened is equal to or greater than a reference number of times.

(9) According to the present disclosure, when it is determined that it is dangerous for an occupant to get off the vehicle in a situation where the open-door detection device detects that a door is open,

the notification of an alert is started and when it is determined that it is no longer dangerous for the occupant to get off the vehicle, the notification of the alert is ended. Further, when a number of times of the alert notified by the notification device after the door is finally opened is equal to or greater than a reference number of times, the alert notified by the notification device is changed so as to reduce annoyance of the alert given to the occupant or occupants.

(10) Therefore, when it is dangerous for the occupant to get off the vehicle, it is possible to inform the occupant who is about to get off the vehicle of the danger by notifying the alert by the notification device. Furthermore, when the number of times of the alert notified after a door is finally opened is equal to or greater than the reference number of times, the annoyance of the alert given to the occupant or occupants is reduced, so that it is possible to reduce the possibility that the occupant or occupants will feel annoyed by the alert.

(11) In one aspect of the present disclosure, the electronic control unit is further configured to start the notification of the alert by the notification device when, in a situation where the open-door detection device detects that a door is open, another vehicle approaching from behind or stopping on the same side as the open door is detected based on the information acquired by the surrounding information acquisition device and end the notification of the alert by the notification device when it is detected that the other vehicle has moved forward from the side of the own vehicle based on the information acquired by the surrounding information acquisition device.

(12) According to the above configuration, the notification of the alert can be started when, in a situation where the open-door detection device detects that a door is open, another vehicle is approaching from behind or stopped on the same side as the open door is detected, and the notification of the alert can be ended when the other vehicle has moved forward from the side of the own vehicle. Therefore, in a situation where it is dangerous for the occupant to get off the vehicle due to another vehicle approaching or stopping, an alert can be notified.

(13) In another aspect of the present disclosure, the electronic control unit is further configured to change the alert notified by the notification device so as to reduce annoyance of the alert given to the occupant or occupants by reducing an appeal of the alert to the occupant or occupants.

(14) In general, the higher the appeal of the alert to the occupant or occupants, the higher the effect of alerting the occupant or occupants, but the more likely the occupant or occupants will feel annoyed by the alert. According to the above aspect, the alert notified by the notification device is changed so as to reduce the annoyance of the alert given to the occupant or occupants by reducing the appeal of the alert to the occupant or occupants.

(15) In another aspect of the present disclosure, the alert notified by the notification device includes multiple types of alarms, and the electronic control unit is further configured to reduce the appeal of the alert to the occupant or occupants by reducing the number of alarms.

(16) In general, the greater the number of types of alarms included in the alert is, the higher the appeal of the alert to the occupant or occupants is. According to the above aspect, by reducing the number of alarms, the appeal of the alert to the occupant or occupants is reduced. Therefore, before the number of times of the alert notified exceeds the reference number of times, all types of alarms are used to alert the occupant or occupants to the necessary alert, and after the number of times of alert notified reaches or exceeds the reference number of times, the appeal of the alert is reduced by reducing the number of alarms, so that it is possible to reduce the possibility that the occupant or occupants will feel annoyed by the alert.

(17) Further, in another aspect of the present disclosure, the multiple types of alarms include an auditory alarm of voice utterance, and the electronic control unit is configured to reduce the appeal of the alert to the occupant or occupants by excluding the auditory alarm of voice utterance.

(18) In general, the auditory alarm of voice utterance is more appealing to the occupant or occupants than an auditory alarm of non-voice utterance and visual alarm. According to the above aspect, the appeal of the alert to the occupant or occupants is reduced by excluding the auditory alarm of voice utterance. Therefore, before the number of times of the alert notified exceeds the

reference number of times, all types of alarms are used to alert the occupant or occupants to the necessary alert, and after the number of times of alert notified reaches or exceeds the reference number of times, the appeal of the alert is reduced by excluding the auditory alarm of voice utterance, so that it is possible to reduce the possibility that the occupant or occupants will feel annoyed by the alert. Furthermore, the necessary alert can be more effectively provided than when all the auditory alarms are excluded.

(19) Further, in another aspect of the present disclosure, the multiple types of alarms include at least one auditory alarm and at least one visual alarm, and the electronic control unit is configured to reduce the appeal of the alert to the occupant or occupants by excluding the auditory alarm.

(20) In general, an auditory alarm is more appealing to the occupant or occupants than a visual alarm. According to the above aspect, by excluding the auditory alarm, the appeal of the alert to the occupant or occupants is reduced. Therefore, before the number of times of the alert notified exceeds the reference number of times, all types of alarms are used to alert the occupant or occupants to the necessary alert, and after the number of times of the alert notified reaches or exceeds the reference number of times, the appeal of the alert is reduced by excluding the auditory alarm, so that it is possible to reduce the possibility that the occupant or occupants will feel annoyed by the alert.

(21) Notably, in the present application, the number of times of the alert notifications is counted as one from the start of the alert notification to the end of the alert notification in a situation where it is detected that a door is open. Further, the reference number of times may be a positive constant integer.

(22) Other objects, other features and attendant advantages of the present disclosure will be readily understood from the description of the embodiments of the present disclosure described with reference to the following drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic configuration diagram showing a first embodiment of a getting-off assistance device according to the present disclosure.

(2) FIG. **2** is a flow chart showing a getting-off assistance control routine in the first embodiment.

(3) FIG. **3** is a flow chart showing a getting-off assistance control routine in the second embodiment.

(4) FIG. **4** shows a situation in which another vehicle is approaching from the right rear of a own vehicle, and neither the right front door nor the right rear door is open.

(5) FIG. **5** shows a situation in which another vehicle is approaching from the right rear of the own vehicle and the right front door is slightly open.

(6) FIG. **6** shows a situation in which the other vehicle has passed the right side of the own vehicle and moved to the right front of the own vehicle.

(7) FIG. **7** shows a situation in which the right front door is slightly open and another vehicle is approaching from the right rear of the own vehicle.

(8) FIG. **8** shows a situation where there is an immovable object on the left side of the vehicle, such as a ditch without lids, which makes getting off the vehicle dangerous.

DETAILED DESCRIPTION

(9) The present disclosure will now be described in detail with reference to the accompanying drawings.

First Embodiment

(10) As shown in FIG. **1**, the getting-off assistance device **100** according to the first embodiment is adapted to a vehicle **102** and includes a vehicle control ECU **10**. The vehicle **102** may be a vehicle

capable of autonomous driving, and includes a body ECU **20**, a meter ECU **30**, and a multimedia ECU **40**. The ECU means an electronic control unit having a microcomputer as its main part. In the following description, the vehicle **102** will be referred to as own vehicle **102** as necessary to distinguish it from other vehicles.

(11) Each microcomputer includes a CPU, ROM, RAM, non-volatile memory (N/M) and an interface (I/F). The CPU realizes various functions by executing instructions (programs, routines) stored in the ROM. Further, these ECUs are connected to each other via a CAN (Controller Area Network) **52** so as to be able to exchange data and communicate. Therefore, detected values of sensors (including switches) connected to a specific ECU are transmitted to other ECUs as well.

(12) The vehicle control ECU **10** is connected with a plurality of camera sensors **12**F, **12**B, **12**R and **12**L and a plurality of radar sensors **14**F, **14**B, **14**R and **14**L. The plurality of camera sensors **12**F, **12**B, **12**R and **12**L are referred to as “camera sensors **12**” when there is no need to distinguish between them. Similarly, the plurality of radar sensors **14**F, **14**B, **14**R and **14**L are referred to as “radar sensors **14**” when there is no need to distinguish between them. The camera sensors **12** and the radar sensors **14** function as a surrounding information acquisition device that acquires information around the vehicle **102**.

(13) Each camera sensor **12**, although not shown in the figure, comprises a camera unit and a recognition unit that analyzes image data obtained by photographing by the camera unit and recognizes targets such as road white lines and other vehicles. The recognition unit of each camera sensor **12** supplies information about the recognized target to the vehicle control ECU **10** every time a predetermined time elapses.

(14) The camera sensor **12**F is a front camera sensor that captures an image of the front of the vehicle **102**, and the camera sensor **12**B is a back camera that captures an image of the rear of the vehicle **102**. The camera sensor **12**R is a right side camera that captures an image of the right side of the vehicle **102**, and the camera sensor **12**L is a left side camera that captures an image of the left side of the vehicle **102**.

(15) Each radar sensor **14** includes a radar transmitting/receiving unit and a signal processor (not shown). The radar transmitting/receiving unit radiates radio waves in a millimeter wave band (hereinafter referred to as “millimeter waves”) and receives millimeter waves (that is, reflected waves) reflected by three-dimensional objects (eg, other vehicles, bicycles, guardrails, etc.) existing within the radiation range. The signal processing unit supplies information representing a distance between the vehicle and the three-dimensional object (hereinafter referred to as surrounding information) to the vehicle control ECU **10** every time a predetermined time elapses based on a phase difference between the transmitted millimeter wave and the received reflected wave, an attenuation level of the reflected wave, and a time period from when the millimeter wave is transmitted to when the reflected wave is received. LiDAR (Light Detection And Ranging) may be used instead of the radar sensor **14**.

(16) The radar sensor **14**F is provided at the front end of the vehicle **102** and acquires surrounding information in front of the vehicle. The radar sensor **14**B is provided at the rear end portion of the vehicle **102** and acquires surrounding information behind the vehicle. The radar sensor **14**R is provided on the right side of the vehicle **102** and acquires surrounding information on the right side of the vehicle. The radar sensor **14**L is provided on the left side of the vehicle **102** and acquires surrounding information on the left side of the vehicle.

(17) Furthermore, an in-cabin camera **16** is connected to the vehicle control ECU **10**. Although not shown in FIG. **1**, the in-cabin camera **16** is attached to a rearview mirror or a vehicle body in the vicinity thereof, and photographs an interior of the vehicle while swinging an optical axis to the left and right. The in-cabin camera **16** supplies the vehicle control ECU **10** with image information obtained by photographing, particularly information regarding the presence or absence of an occupant. If the vehicle **102** is equipped with a drive recorder device having a camera that captures 360-degree images of the exterior and interior of the vehicle, the in-cabin camera **16** may be the

camera of the drive recorder device.
(18) A plurality of door sensors **22**FR, **22**FL, **22**RR and **22**RL are connected to the body ECU **20**. The door sensors **22**FR, **22**FL, **22**RR and **22**RL function as door opening detectors for detecting opening of a right front door **24**FR, a left front door **24**FR, a right rear door **24**RR and a left rear door **24**RL, respectively. The plurality of door sensors **22**FR, **22**FL, **22**RR and **22**RL are referred to as “door sensors **22**” when there is no need to distinguish between them. The doors **24**FR, **24**FR, **24**RR and **24**RL are referred to as “doors **24**” when there is no need to distinguish between them. Each door sensor **22** outputs a signal indicating that the corresponding door **24** is opened to the body ECU **20** and to the vehicle control ECU **10** via the CAN **52**, when the corresponding door **24** is opened.
(19) Indicators **32**R and **32**L and a buzzer **34** are connected to the meter ECU **30**. The indicators **32**R and **32**L are installed on the right and left door mirrors, respectively, so as to be visible from inside the vehicle. The indicators **32**R and **32**L illuminate or blink when an actuation signal is output from the meter ECU **30** in response to an alert command transmitted from the vehicle control ECU **10**, and issue a visual alarm. Indicators **32**R and **32**L are referred to as “indicators **32**” when there is no need to distinguish between them. Further, the buzzer **34** sounds when a buzzer sounding signal is output from the meter ECU **30** in response to an alarm command transmitted from the vehicle control ECU **10**, and issues an auditory alarm.
(20) A display **42** and a speaker **44** are connected to the multimedia ECU **40**. The multimedia ECU **40** displays an image on the display **42** according to a display command transmitted from the vehicle control ECU **10**. The multimedia ECU **40** also causes the speaker **44** to issue a voice utterance alarm in accordance with a voice utterance command transmitted from the vehicle control ECU **10**. Thus, display **42** provides a visual alert and speaker **44** provides an audible alert. Note that the speaker **44** may be, for example, a speaker of an audio device. Further, the display **42** may be, for example, a head-up display or a multi-information display that displays meters and various information, or may be a display of a navigation device.
(21) The visual alarm displayed on the display **42** is preferably a visual alarm that gives specific dangerous information, such as “a vehicle is approaching from the right rear” or “there is a ditch on the left side of the vehicle”. Also, the voice utterance alarm generated by the speaker **44** is preferably an audible alarm that notifies specific dangerous information such as “a vehicle is approaching from the right rear” or “there is an open ditch on the left side of the vehicle”. It should be noted that the above voice utterance alarm generated by the speaker **44** is repeatedly notified at regular time intervals.
(22) As can be seen from the above description, the indicator **32**, buzzer **34**, display **42** and speaker **44** function as an alert device **50** that issues visual and/or audible alarms. Although not shown in FIG. **1**, an ECU such as the vehicle control ECU **10**, the sensors such as the camera sensors **12**, and the alert device **50** continue to be energized for a preset operation time (positive constant) from a time an ignition switch is turned off.
(23) In the first embodiment, the ROM of the vehicle control ECU **10** stores a getting-off assistance control program corresponding to the flowchart shown in FIG. **2** and the CPU executes the getting-off assistance control according to the program so that an occupant or occupants can safely get off the vehicle. That is, when the vehicle control ECU **10** determines that it is dangerous for the occupant or occupants to get off the vehicle based on the information around the vehicle **102** acquired by the camera sensors **12** and the radar sensors **14** and detects that a door is open, all the alarm devices such as the buzzer **34** are activated to issue all the alarms. The vehicle control ECU **10** may perform various vehicle controls known in the art, such as lane departure prevention control and collision damage mitigation control, while the vehicle **102** is running.
Getting-Off Assistance Control Routine in First Embodiment
(24) Next, the getting-off assistance control routine in the first embodiment will be described with reference to the flowchart shown in FIG. **2**. The getting-off assistance control according to the

flowchart shown in FIG. **2** is started by the CPU of the vehicle control ECU **10** when an ignition switch (not shown in FIG. **1**) is turned off. In addition, the getting-off assistance control is repeatedly executed at a predetermined control cycle in order of the surrounding conditions on the right side of the vehicle **102** and the right front door **24**FR, the surrounding conditions on the left side of the vehicle and the left front door **24**FL, the surrounding conditions on the right side of the vehicle and the right rear door **24**RR, the surrounding conditions on the left side of the vehicle and the left rear door **24**RL. In the following description, the getting-off assistance control is simply referred to as "present control".

(25) Note that "x" of flags Fax and Fbx and a count value Nx of a counter, which will be described later, is "fr" when the getting-off assistance control is being executed for the surrounding situation on the right side of the vehicle **102** and the right front door **24**FR and is "fl" when the control is being executed for the surrounding situation on the left side of the vehicle and the left front door **24**FL. Note that "x" is "rr" when the getting-off assistance control is being executed for the surrounding situation on the right side of the vehicle **102** and the right rear door **24**RR and is "rl" when the control is being executed for the surrounding situation on the left side of the vehicle and the left rear door **24**RL. At the start of the getting-off assistance control, the flags Fax and Fbx and the count value Nx of the counter are reset to 0, respectively. These are the same for the second embodiment (FIG. **3**) described later.

(26) First, in step S**10**, the CPU determines whether or not the flag Fax is 1, that is, determines whether or not an alert has already been notified by the alert device **50** by executing step S**120** or S**130** described later. When the CPU makes an affirmative determination, the present control proceeds to step S**140**, and when it makes a negative determination, the present control proceeds to step S**20**.

(27) In step S**20**, the CPU determines whether or not the camera sensor **12** and/or the radar sensor **14** detect a dangerous situation that occurs if an occupant gets off the vehicle. In this connection, as a dangerous situation, for example, it is determined on the right side of the vehicle **102** whether or not an object, such as another vehicle or a wild animal that may pass close to the vehicle **102** from the right rear or right front of the vehicle is detected. it is determined for the left side of the vehicle **102** whether or not an object that may pass close to the vehicle **102** from the left rear or left front of the vehicle, such as another vehicle such as a bicycle or motorcycle, or a wild animal is detected and whether a dangerous object, such as an uncovered ditch, is detected. When the CPU makes a negative determination, it once ends the present control, and when it makes an affirmative determination, it advances the present control to step S**30**.

(28) In step S**30**, when the CPU determines in step S**20** that a dangerous situation is detected on the right side of vehicle **102**, it turns on the right indicator **32**R. On the other hand, when the CPU determines in step S**20** that a dangerous situation is detected on the left side of the vehicle **102**, it turns on the left indicator **32**L. When dangerous situations are detected on the right and left sides of the vehicle **102**, the right and left indicators **32**R and **32**L are turned on.

(29) In step S**40**, the CPU determines whether or not the corresponding door sensor **22** detects that the corresponding door **24** is open. When the CPU makes a negative determination, it advances the present control to step S**50**, and when it makes an affirmative determination, it advances the present control to step S**60**.

(30) In step S**50**, the CPU resets the count value Nx of the counter to 0, and then temporarily terminates the present control.

(31) In step S**60**, the CPU determines whether or not the determination in step S**20** has changed from a negative determination to an affirmative determination while the corresponding door sensor **22** detects that the corresponding door **24** is open. When the CPU makes a negative determination, the present control advances to step S**110**, and when it makes an affirmative determination, that is, when a dangerous situation that occurs if an occupant gets off the vehicle changes from not being detected to being detected, the count value Nx of the counter is incremented by one in step S**70**.

(32) In step S**80**, the CPU determines whether or not the count value Nx of the counter is greater than or equal to a reference count Nxc. When the CPU makes a affirmative determination, it sets the flag Fbx to 1 in step S**90**, and when the CPU makes a negative determination, it resets the flag Fbx to 0 in step S**100**. Note that the reference count Nxc may be a constant positive integer.
(33) In step S**110**, the CPU determines whether or not the flag Fbx is 1, ie, whether or not voice utterance alarm should be excluded from the alert. When the CPU makes an affirmative determination, the present control proceeds to step S**130**, and when the CPU makes a negative determination, the present control proceeds to step S**120**.
(34) In step S**120**, the CPU outputs command signals to the meter ECU **30** and the multimedia ECU **40** so that all the alarms including voice utterance alarm are issued. In this step, the indicator **32** is changed from lighting to blinking. Also, the CPU sets the flag Fax to one.
(35) In a situation where one of the front door and the rear door on the same side of the vehicle **102** is already open, when the other of the front door and the rear door is opened, since all the alarms or the alarms excluding the voice utterance alarm are already issued, only the flag Fax is set to 1 for the other door.
(36) In step S**130**, the CPU outputs command signals to the meter ECU **30** and the multimedia ECU **40** so that all the alarms other than the voice utterance alarm are issued. Also in this step, the corresponding indicator **32** is changed from lighting to blinking. Also, the CPU sets the flag Fax to one.
(37) Note that when all the alarms have been issued for both the left and right sides of the vehicle **102**, the voice utterance alarm is excluded only for the side for which the affirmative determination was made in step S**80**. However, at this timing, the voice utterance alarm may be excluded for both the left and right sides of the vehicle **102**.
(38) In step S**140**, the CPU determines whether the door sensor **22** has detected that the corresponding door **24** is closed. When the CPU makes a negative determination, it advances the present control to step S**160**, and when the CPU makes an affirmative determination, it resets the flag Fbx and the count value Nx of the counter to 0 in step S**150**.
(39) In step S**160**, the CPU determines whether or not an alert end condition other than closing of the door **24** is satisfied. When the CPU makes a negative determination, it once ends the present control, and when the CPU makes an affirmative determination, it advances the present control to step S**170**. In this connection, when any one of the following conditions E1 to E3 is satisfied, it is determined that the alert end condition other than closing of the door is satisfied. E1: The camera sensor **12** and the radar sensor **14** stopped detecting a dangerous situation generated when an occupant got off the vehicle. E2: There is no occupant detected by the in-cabin camera **16**. That is, all occupants got off the vehicle. E3: A preset control time (positive constant) has elapsed since the alert was initiated.
(40) In step S**170**, the CPU terminates issuing of all the alarms and resets the flag Fax to zero. When the condition E2 or E3 is satisfied, the power supply to the ECUs such as the vehicle control ECU **10**, the sensors such as the camera sensors **12**, and the alert device **50** may be terminated.
(41) In addition, when one of the front door and the rear door is closed in a situation where both the front door and the rear door on the same side of the vehicle **102** are open, the alert notification is maintained, and the flag Fbx and the count value Nx of the counter may be reset for the one of the doors.
Example of Operation of First Embodiment
(42) Next, the operation of the first embodiment will be described with reference to FIGS. **4** to **6**, taking as an example a case where another vehicle approaches from the right rear of the vehicle **102** and passes on the right side.
(43) FIG. **4** shows a situation in which another vehicle **60** is approaching from the right rear of the own vehicle **102**, and neither the right front door **24**FR nor the right rear door **24**RR is open. In this situation, a negative determination is made in step S**10** and an affirmative determination is made in

step S**20**, so that right indicator **32**R is lit up in step S**30**. Further, a negative determination is made in step S**40**. Therefore, an occupant or occupants can recognize the danger of opening the right side door or doors and getting off the vehicle because the indicator **32**R is lit.

(44) FIG. **5** shows a situation in which another vehicle **60** is approaching from the right rear of the own vehicle **102** and the right front door **24**FR is slightly opened. In this situation, affirmative determinations are made in steps S**20**, S**40** and S**60**, so that the count value Nx of the counter is counted up to 1 in step S**70**. Assuming that the reference count Nxc is 2, a negative determination is made in step S**80**, and the flag Fbx is reset to 0 in step S**100**. Therefore, since a negative determination is made in step S**110**, all the alarms including the voice utterance alarm are issued in step S**120**. Thus, a driver can stop opening the front door **24**FR any further to get off the vehicle, thereby avoiding the danger of the other vehicle **60** colliding with the driver and the front door **24**FR.

(45) In this situation, when the right front door **24**FR is closed, affirmative determinations are made in steps S**10** and S**140**, so that in S**170**, the issuing of all the alarms is terminated, and the flag Fax and the like are reset to zero. Therefore, the occupant or occupants do not feel annoyed by the alert.

(46) FIG. **6** shows a situation in which the right front door **24**FR remains slightly open, and the other vehicle **60** passes the right side of the own vehicle **102** and moves to the right front of the own vehicle **102**. In this situation, when the other vehicle **60** passes on the right side of the own vehicle **102** and the condition E1 is satisfied, that is, when the camera sensors **12** and the radar sensors **14** no longer detect a dangerous situation even if the occupant gets out of the vehicle, the determination in S**160** becomes an affirmative determination. Therefore, even if the right front door **24**FR is closed, all the alarms are terminated so that the driver can get out of the vehicle with security.

(47) FIG. **7** shows a situation in which the right front door **24**FR remains slightly open and another vehicle **62** is approaching the own vehicle **102** from the right rear. In this situation, as in the case shown in FIG. **5**, affirmative determinations are made in steps S**20**, S**40** and S**60**, so the count value Nx of the counter is counted up to 2 in step S**70**. Assuming that the reference count Nxc is 2, an affirmative determination is made in step S**80**, and the flag Fbx is set to 1 in step S**90**. Therefore, since an affirmative determination is made in step S**110**, the alarms other than the voice utterance alarm are issued in step S**130**. Thus, the driver can stop opening the front door **24**FR any further to get off the vehicle, thereby avoiding the risk of the other vehicle **60** colliding with the driver and the front door **24**FR, and furthermore, it is possible to reduce a possibility that the occupant or occupants will feel annoyed by the alert.

(48) Although not shown in the drawings, even when another vehicle such as a motorbike passes on the left side of the own vehicle **102**, the getting-off assist device **100** works the same as that shown in FIGS. **4** to **7**.

(49) In addition, as shown in FIG. **8**, when there is an immovable object such as a side ditch **64** without lids on the left side of the vehicle **102**, which makes it dangerous to get off the vehicle, the condition E1 is not satisfied so that the alert end condition is not satisfied. Therefore, when an immobile object that makes getting off the vehicle dangerous is detected in step S**20**, the alert issued in steps S**120** and S**130** may include information recommending that the occupant or occupants get off the vehicle from the left side of the vehicle at another safe place. Incidentally, in the case shown in FIG. **8**, the condition for terminating the alarms is satisfied when the condition E3 is satisfied, so that the alert does not continue for an excessively long time.

Second Embodiment

(50) The getting-off assistance device **100** according to the second embodiment of the present invention is configured in the same manner as the getting-off assistance device according to the first embodiment, and the CPU of the vehicle control ECU **10** performs the getting-off assistance control according to the flowchart shown in FIG. **3**.

Getting-Off Assistance Control Routine in Second Embodiment

(51) Next, the getting-off assistance control routine in the second embodiment will be described with reference to the flowchart shown in FIG. **3**. In FIG. **3**, steps that are the same as those shown in FIG. **2** are given the same step numbers as the step numbers given in FIG. **2**.
(52) In the second embodiment, in step S**10**, the CPU determines whether or not the flag Fax is 1 as in the first embodiment. When an affirmative determination is made, the present control proceeds to step S**140**, and when a negative determination is made, the present control proceeds to step S**25**.
(53) In step S**25**, the CPU determines whether or not opening of the corresponding door **24** is detected by the corresponding door sensor **22**, as in step S**40** in the first embodiment. When the CPU makes a negative determination, it advances the present control to step S**50**, and when the CPU makes an affirmative determination, it advances the present control to step S**45**.
(54) In step S**45**, the CPU determines whether or not the camera sensor **12** and/or the radar sensor **14** detect a dangerous situation that occurs if the occupant gets out of the vehicle, as in step S**20** in the first embodiment. When the CPU makes a negative determination, it once ends the present control, and when the CPU makes an affirmative determination, it advances the present control to step S**60**.
(55) As can be seen from a comparison of FIGS. **3** and **2**, steps S**50** through S**170** are performed in the same manner as steps S**50** through S**170** in the first embodiment, respectively. Therefore, description of these steps is omitted.
(56) In the second embodiment, step S**30** of the first embodiment is not executed. Therefore, the getting-off assistance device of the second embodiment works the same as the getting-off assistance device of the first embodiment, except that the indicator **32** is not lit even if a dangerous situation that occurs when the occupant gets off the vehicle is detected by the camera sensors **12** and/or the radar sensors **14**. Note that the indicator **32** may be lit or blinked in steps S**120** and S**130**.
(57) As can be seen from the above description, according to the first and second embodiments, when it is determined that it is dangerous for an occupant to get off the vehicle (S**20**, S**45**) and opening of a door is detected (S**40**, S**25**), the alert device **50** such as the indicator **32** issues alarms including an auditory alarm of voice utterance (S**120**). However, when the number of times Nx of alert notifications reaches or exceeds the reference number of times Nxc (S**80**), the voice utterance alarm is excluded from a plurality of types of alarms to be issued, and the degree of appeal of the alert is reduced, whereby the alert notified by the alert device **50** is changed so as to reduce the annoyance of the alert given to the occupants (S**130**).
(58) Therefore, before the number of times Nx of alarm notifications reaches or exceeds the reference number of times Nxc, it is possible to ensure the appeal of the alert by the alarms including the auditory alarm of voice utterance to a necessary level and necessary alerting can be performed to the occupant or occupants. In addition, after the number of times Nx of alert notifications has reached or exceeded the reference number of times Nxc, since the appeal of the alert is reduced by excluding the voice utterance alarm, it is possible to reduce the possibility that the occupant or occupants will feel annoyed by the alert.
(59) In addition, according to the first and second embodiments, when the number of times Nx of alert notifications reaches or exceeds the reference number of times Nxc, the alarm of voice utterance is excluded from the plurality of types of alarms to be issued. Therefore, by reducing the number of alarms, excluding audio utterance alarm with high alerting appeal among the auditory alarms, and excluding one auditory alarm, the appeal of the alert can be effectively reduced.
(60) Furthermore, according to the first and second embodiments, when the number of times Nx of alert notifications reaches or exceeds the reference number of times Nxc, the voice utterance alarm is excluded from the plurality of types of alarms to be issued, but an audible alarm by the buzzer **34** which is a non-voice utterance auditory alarm is not excluded. Therefore, compared to where all auditory alarms are excluded, necessary alerting can be performed effectively.
(61) In particular, according to the first embodiment, it is first determined whether or not a dangerous situation that occurs if the occupant gets off the vehicle is detected (S**20**), and when a

dangerous situation is detected, the indicator **32** is turned on. Therefore, before the occupant opens the door **24**, he or she can be alerted that there is a dangerous situation if he or she gets off the vehicle.

(62) Although the present disclosure has been described in detail with reference to specific embodiments, it will be apparent to those skilled in the art that the present disclosure is not limited to the above-described embodiments, and various other embodiments are possible within the scope of the present disclosure.

(63) For example, in the above-described first and second embodiments, the alarms include an auditory alarm of voice utterance, and when the number of times Nx of alerts reaches or exceeds the reference number of times Nxc, the alarm of voice utterance is excluded, thereby reducing the appeal of the alert. However, the reduction in appeal of the alert may be achieved by increasing the time interval between voice utterances, or the appeal of the alert may be reduced by excluding all auditory alarms. In addition, the auditory alarms may not include the auditory alarm of voice utterance. In that case, when the number of alert notifications Nx reaches or exceeds the reference number of times Nxc, the appeal of the alert may be reduced by excluding audible alarms.

(64) Further, in the above-described first and second embodiments, the appeal of the alert is also reduced by reducing the number of alarms by excluding voice utterance alarm. However, the appeal of the alert may be reduced by changing the mode of one or more types of alarms. For example, for an auditory alarm, the appeal of the alert may be reduced by reducing a volume, lowering a frequency, etc. In particular, for the alarm issued by the buzzer **34**, in addition to the above manners, the appeal of the alert may be reduced by increasing an intermittent time interval of the buzzer sound. As for the visual alarm, the appeal of the alert may be reduced by decreasing a size of a text alarm, increasing a display time interval of the text alarm, or decreasing a brightness or luminance of the alarm.

(65) Further, in the above-described first and second embodiments, determination as to whether or not the alert termination condition E2 is satisfied is performed based on image information acquired by photographing by the in-cabin camera **16**. However, in a vehicle in which each seat is provided with a seat sensor, the determination as to whether the condition E2 is satisfied may be made based on the detection results of the seat sensors.

(66) Further, in the above-described first and second embodiments, the alerts include visual and audible alerts, but one of the visual and audible alerts may be omitted. Conversely, the alerts may include tactile alerts, such as seat vibrations, in addition to visual and audible alerts.

(67) Further, in the above-described first and second embodiments, when it is determined in step S**160** that the condition E3 is satisfied, all the alarms are terminated in step S**170**. However, when it is determined in step S**160** that the first control time (positive constant) has elapsed from a time point when the alert was started, the control may proceed to step S**130** where voice utterance alarm is excluded, and when it is determined in step S**160** that the second control time (positive constant greater than the first control time) has elapsed from the time point when the alert was started, the control may proceed to step S**170**. According to this modification, it is possible to reduce the possibility that an occupant or occupants may feel annoyed by the alert even when a plurality of objects sequentially approach and pass through the own vehicle without leaving an interval.

(68) Further, in the above-described first and second embodiments, the getting-off assistance control according to the flowcharts shown in FIGS. **2** and **3** is started by the CPU of the vehicle control ECU **10** when the ignition switch is turned off. However, the getting-off assistance control according to the flowcharts shown in FIGS. **2** and **3** may also be executed when the ignition switch is on and the vehicle is stopped.

(69) Further, in the above-described first embodiment, when it is determined in step S**20** that a dangerous situation is detected when an occupant gets off the vehicle, the indicator **32** is turned on in step S**30**. However, step S**30** may be omitted.

Claims

1. A getting-off assistance device for a vehicle that includes a surrounding information acquisition device that acquires information about surroundings of an own vehicle, a notification device that notifies an alert to an occupant or occupants of the own vehicle, an open-door detection device that detects opening of a door, and an electronic control unit that controls the notification device, and the electronic control unit is configured to start notification of the alert by the notification device when it is determined that it is dangerous for the occupant or occupants to get off the vehicle based on the information acquired by the surrounding information acquisition device in a situation where the open-door detection device detects that a door is open, and end the notification of the alert by the notification device when it is determined that it is no longer dangerous for the occupant or occupants to get off the vehicle based on the information acquired by the surrounding information acquisition device, wherein the electronic control unit is further configured to change the alert notified by the notification device so as to reduce annoyance of the alert given to the occupant or occupants when a number of times of the alerts notified by the notification device during a situation where the door is open after the door is finally opened is equal to or greater than a reference number of times, and the electronic control unit is further configured to terminate the notification of the alert by the notification device and reset the number of times of the alerts to zero when the door-open detection device detects that the door has been closed.
2. The getting-off assistance device for a vehicle according to claim 1, wherein the electronic control unit is further configured to start the notification of the alert by the notification device when, in a situation where the open-door detection device detects that a door is open, another vehicle approaching from behind or stopping on the same side as the open door is detected based on the information acquired by the surrounding information acquisition device and end the notification of the alert by the notification device when it is detected that an other vehicle has moved forward from the side of the own vehicle based on the information acquired by the surrounding information acquisition device.
3. The getting-off assistance device for a vehicle according to claim 1, wherein the electronic control unit is further configured to change the alert notified by the notification device so as to reduce annoyance of the alert given to the occupant or occupants by reducing an appeal of the alert to the occupant or occupants.
4. The getting-off assistance device for a vehicle according to claim 3, wherein the alert notified by the notification device includes multiple types of alarms, and the electronic control unit is further configured to reduce the appeal of the alert to the occupant or occupants by reducing a number of alarms.
5. The getting-off assistance device for a vehicle according to claim 4, wherein the multiple types of alarms include an auditory alarm of voice utterance, and the electronic control unit is configured to reduce the appeal of the alert to the occupant or occupants by excluding the auditory alarm of voice utterance.
6. The getting-off assistance device for a vehicle according to claim 4, wherein the multiple types of alarms include at least one auditory alarm and at least one visual alarm, and the electronic control unit is configured to reduce the appeal of the alert to the occupant or occupants by excluding the auditory alarm.